

Kneser graphs $K(2k + n, k)$ with $n > 0$ are not CFON* 1-colorable:
the cases $K(2k + 1, k)$ and $K(2k + 2, k)$

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Let K be the Kneser graph being considered, and let T be an efficient open dominating set of K , the set of all colored vertices in a CFON* 1-coloring.

Let each vertex, v , in K be assigned a variable:

$$x_v = \begin{cases} 1, & v \in T \\ 0, & v \notin T \end{cases}$$

Let A be the adjacency matrix, with each row and column representing a vertex, and elements where corresponding vertices for the row-column pair are adjacent equals 1, the rest of the elements equal 0. Let x be the vector of the values x_v . Each element of Ax is the number of colored vertices in the neighborhood of the corresponding vertex, so if T is an efficient open dominating set then

$$Ax = \mathbf{1},$$

where $\mathbf{1}$ is the vector with every element equal to 1.

1 The odd graph $K(2k + 1, k)$

The odd graph $K(2k + 1, k)$ has no efficient open dominating set.

All Kneser graphs are regular, so every vertex has the same degree. For our $K(2k + 1, k)$ graph:

$$\binom{2k + 1 - k}{k} = \binom{k + 1}{k} = \frac{(k + 1)!}{1! k!} = k + 1$$

If we have sets of size k , then we have $(2k + 1) - k$ sets to form adjacent vertices with.

Thus, each row of the adjacency matrix has $k + 1$ 1's. We can try to solve the matrix equation, ignoring coloring rules for now.

→ We want every value in the product vector to be a 1, so every value of x has to be the same.

$$A \cdot x = A \cdot \begin{bmatrix} c \\ c \\ \vdots \\ c \end{bmatrix} = \mathbf{1}$$

When we multiply each row of A , we get:

$$\underbrace{c + c + c + \cdots + c}_{k+1} = (k+1) \cdot c$$

Each row has to equal 1, so:

$$(k+1) \cdot c = 1, \quad c = \frac{1}{k+1}$$

So

$$x = \begin{bmatrix} 1/(k+1) \\ 1/(k+1) \\ \vdots \\ 1/(k+1) \end{bmatrix} = \frac{1}{k+1} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Now we can find the eigenvalues of our $Ax = \mathbf{1}$ matrix using the odd graph formula

$$\lambda_j = (-1)^j (k+1-j), \quad j = 0, 1, \dots, k$$

The range of magnitudes for our eigenvalue is from 1 ($k = j$) to $k+1$ ($0 = j$). 0 is not in this range, so the matrix is invertible.

Now suppose there was some vector w such that $Aw = \mathbf{1}$, and $w \in \mathbb{N}_0^{|V(K)|}$ such that it meets CFON* coloring restrictions.

Then:

$$\begin{aligned} Aw &= Ax = \mathbf{1} \\ A^{-1}Aw &= A^{-1}Ax && \text{(we've proven invertibility)} \\ w &= x, \end{aligned}$$

but x is fractional which contradicts the definition of w . Thus, w cannot exist, and x is the unique solution.

Since x is the unique solution and its elements are fractional, and since x_v can only equal 0 or 1 based on its inclusion in T , no efficient open dominating set exists and thus $K(2k+1, k)$ is not 1-colorable.

Thus, $K(29, 14)$ is not 1-colorable.

2 $K(2k+2, k)$

Now for the existence of an efficient open dominating set for $K(2k+2, k)$.

The $K(2k+2, k)$ graph is $\binom{k+2}{k}$ -regular.

So, we have $\frac{k^2 + 3k + 2}{2}$ degrees for every vertex.

Let B be the adjacency matrix, with each row and column representing a vertex, and elements where corresponding vertices for the row-column pair are adjacent equals 1, the rest of the elements equal 0.

Note again that this matches with the values assigned based on T :

$$x_v = \begin{cases} 1, & v \in T \\ 0, & v \notin T \end{cases}$$

Each row of the adjacency matrix has $\frac{k^2 + 3k + 2}{2}$ 1's.

Then, suppose

$$By = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Suppose all the values of y are some constant d . Then:

$$\left(\frac{k^2 + 3k + 2}{2} \right) d = 1$$

$$d = \frac{2}{k^2 + 3k + 2} < 1, \quad \forall k, k \in \mathbb{Z}^+$$

The eigenvalues for B :

$$\lambda_j = (-1)^j \binom{k+2-j}{2} \quad \text{for } j = 0, 1, \dots, k$$

If $j = 0$, then $\binom{k+2}{2} > \binom{2}{2} = 1 > 0$.

If $j = k$, then $\binom{2}{2} = 1 > 0$.

Thus the eigenvalue for B is always non-zero, making B invertible.

By way of contradiction, assume a matrix p exists where $Bp = \mathbf{1}$, and $p \in \mathbb{N}_0^{|V(K)|}$.

$$Bp = By, \quad B^{-1}Bp = B^{-1}By, \quad p = y,$$

which is a contradiction because y is fractional and p can't be. Thus there is no efficient open dominating set for $K(2k+2, k)$.